



Clinical Practice Guidelines

Adult Hyperosmolar Hyperglycaemic State (HHS)

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy,
Out of Scope	This guideline must NOT be used in: - Children (<16 years) even if they have HHS. Seek specialist advice. - In the management of Diabetic Ketoacidosis (DKA).	

Introduction

This guideline provides hospital specific guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the management of Adult Hyperosmolar Hyperglycaemic State (HHS) within SCGOPHCG. This document supports and complies with related legislation, policy and other related documents [here](#).

General Information

Continuous cardiac monitoring must be instituted when [intravenous potassium chloride](#) is administered at a rate greater than 10mmol/hour.

OPH patients diagnosed with DKA/HHS require early liaison with SCGH Endocrinology. Routine continuous cardiac monitoring for DKA / HHS patients is not available at OPH per the [SCGOPHCG High Risk Medication Policy](#). Patients requiring intravenous potassium administered at a rate greater than 10mmol/hour must be transferred to SCGH.

HHS is a medical emergency and is different from diabetic ketoacidosis (DKA), requiring a different treatment approach¹. HHS typically occurs in the elderly, however, may be seen in younger adults often as the first presentation of type 2 diabetes mellitus¹. It has a high mortality and may be complicated by myocardial infarction, stroke, peripheral arterial thrombosis, seizures, cerebral oedema and central pontine myelinolysis (CPM). There is some evidence that rapid changes in osmolality during treatment may be the precipitant of CPM. Whilst DKA presents within hours of onset, HHS develops over many days, consequently dehydration/ metabolic disturbances are more extreme.¹

1.0 Indications

- This guideline is for the treatment and management of adults with HHS, previously known as hyperosmolar non-ketotic coma (HONK). Seek advice from senior clinician or diabetes specialty unit in cases of uncertainty.

2.0 Guideline information

Staff should adhere to the following guideline when managing an adult patient with Hyperosmolar Hyperglycaemic State (HHS).

The guideline must be adapted to the clinical situation where necessary.

All management must be recorded in Adult Hyperosmolar Hyperglycaemic State Guidelines and Management Record, see [appendix 1](#).

The supplementation of potassium may be via the enteral (e.g. oral, nasogastric, PEG) or intravenous route. Intravenous replacement of potassium should only be prescribed when there is no suitable alternative route following appropriate clinical assessment by a medical officer.

3.0 Diagnosis (Entry) Criteria (Use the Graph on page 5 to assist)

- Hypovolaemia.
- Marked hyperglycaemia (>30 mmol/L).
- Increased plasma osmolality (> 320 mmol/kg or mosmol/kg).
- Calculated osmolality = $2 \text{ (Sodium)} + \text{glucose} + \text{urea}$ (all in mmol/L).
(N.B. osmolality and osmolarity units can be used interchangeably in this clinical scenario).

If the patient does not meet any of the above criteria, re-consider diagnosis.

If the patient has Type 1 diabetes or is hyperglycaemic (blood glucose level >11 mmol/L) with:

- acidosis (venous bicarbonate <18 mmol/L and/or pH <7.35), **and**
- blood (capillary) ketones ≥ 3 mmol/L).

Follow the [SCGH Adult Diabetes Ketoacidosis Guidelines](#).

4.0 Essential Assessment to Determine the Severity of HHS

The following assessment must be carried out to determine the severity of HHS:

- Blood (capillary) ketones.
- Blood glucose level (BGL).
- Venous bicarbonate (HCO_3).
- Venous (or arterial) pH.
- Potassium - beware of initial low potassium; if level <3.5 mmol/L, call for senior medical advice immediately.
- Serum osmolality.
- Serum Creatinine.
- Urea.
- Systolic Blood pressure (SBP).
- Glasgow Coma Scale (GCS).

5.0 Principles for the Management of HHS

1. Correct hypovolaemia first by using 0.9% sodium chloride solution for resuscitation. Do not use Hartmann's or colloid. **Expect an initial rise in sodium level.**
2. Closely monitor electrolytes, as there is a high risk of abnormalities & cerebral oedema.
3. **Treat hyperglycaemia with intravenous (IV) fluids only.**
4. Only commence insulin when:
 - blood glucose is no longer falling with IV fluids, **or**
 - significant ketonaemia - blood (capillary) ketones: **>1 mmol/L.**
5. Secondary risk prevention must commence. Assess foot risk score on admission.

Consider critical care admission and senior clinician review if one or more of the following is present:

- Osmolality >350 mosmol/kg.
- Systolic Blood Pressure (SBP) <90 mmHg.
- Plasma Sodium (Na⁺) > 160 mmol/L.
- Pulse <60 or >100 bpm.
- pH < 7.1
- Hypokalaemia or hyperkalaemia.
- Urine output < 0.5 mL/kg/hr.
- Glasgow Coma Scale (GCS) < 12 or abnormal AVPU score.
- O₂ (oxygen) saturation < 92% on room air.
- Other serious co-morbidity.

6.0 Intravenous Fluid Replacement

Assess severity of dehydration in the patient and use sodium chloride 0.9% (+/- potassium) for fluid replacement WITHOUT insulin initially.

This alone will lower blood glucose which will reduce osmolality.

- Aim to achieve positive fluid balance of 2 – 3 litres within the first 6 hours of intervention, reassess for fluid requirements for the next 6 hours and aim to achieve 100% of estimated fluid losses by 24 hours.
- Typically, fluid losses approximate 110 to 220 mL/kg (i.e. 11L to 22 L for a 100kg patient).
- Provided osmolality is decreasing appropriately, continue patient on sodium chloride 0.9%, **even if plasma sodium is increasing.**
- All fluid, after the first bag, should contain potassium chloride unless urine output is < 0.5 mL/kg/hr or serum potassium remains > 5.5 mmol/L.

Caution is needed, particularly in the elderly, where rapid rehydration may precipitate heart failure.

Table 1: Example fluid regimen (can speed up initial fluids as clinically indicated)

Bag	IV Fluids and Additives	Rate
1 st bag	0.9% sodium chloride 1L	1000 mL over 1 hour
2 nd bag	Sodium chloride 1L with 40 mmol potassium chloride	1000 mL over next 2 hours**
3 rd bag	Sodium chloride 1L with 40 mmol potassium chloride	1000 mL over next 4 hours
4 th bag	Sodium chloride 1L with 40 mmol potassium chloride	1000 mL over next 4 hours
5 th bag	Sodium chloride 1L with 40 mmol potassium chloride	1000 mL over next 6 hours

7.0 Intravenous Potassium Management

- Initial potassium is often high, particularly if patient has renal impairment.
- Caution needs to be taken if urine output is less than 0.5mL/kg/hr.
- Insulin and fluid replacement often cause an acute drop in potassium, particularly within the first 1 to 2 hours.

Rate of [potassium infusion](#) through peripheral lines must be no more than 20mmol potassium chloride per hour in HHS. **The patient must be on a cardiac monitor if receiving > 10mmol/hr potassium chloride, or if serum potassium is > 5.5mmol/L or < 3mmol/L).

Table 2: Medical action according to the patient's potassium level

Parameters	Action
Potassium > 5.5mmol/L	Nil replacement required
Potassium 3.5 to 5.5 mmol/L	20mmol to 40mmol potassium (premixed bags only) **Maximum rate peripherally is 20mmol/hr with cardiac monitoring.
Potassium < 3.5mmol/L	Senior medical review required immediately

8.0 Insulin Management

Only start low dose IV insulin at 0.05 units/kg/hour (e.g. 80kg = 4 units/hr) if:

- a) Significant ketonaemia - bedside blood (capillary) ketones >1 mmol/L.

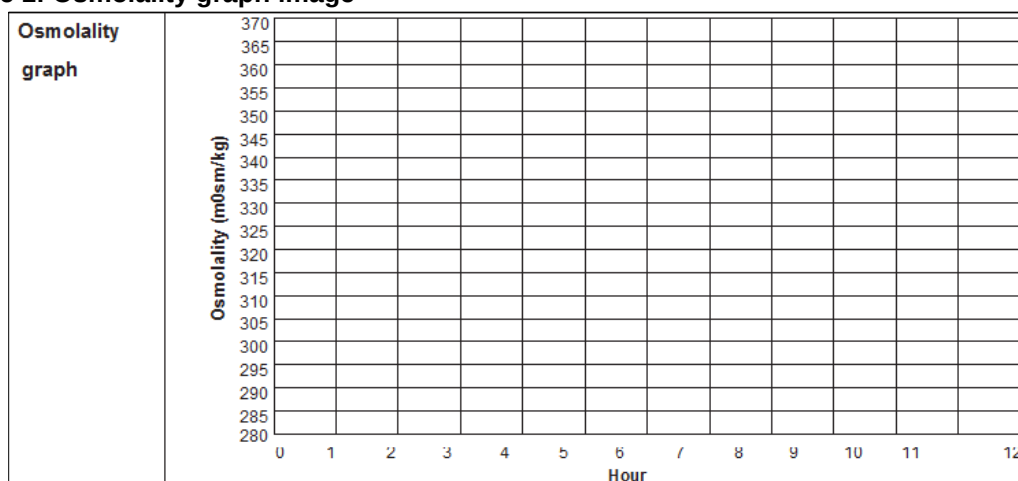
OR

- b) Blood glucose level falling at a rate of less than 5mmol/hr despite adequate fluid replacement.
- Add 50 units of Actrapid® to 50mL of sodium chloride 0.9%.
 - Commence IV insulin rate at 0.05units/kg/hour. Ensure independent pump is used.
 - Once commenced, if blood glucose is falling at a rate less than 5mmol/L per hour with adequate fluid balance, increase the insulin rate to 0.1 units/kg/hr.
 - **Beware of hypoglycaemia.**
 - **Aim to keep blood glucose 10 to 15mmol/L in the first 24 hours.**
 - Once blood glucose falls below 14mmol/L:
 - commence 10% glucose at 125mL/hr via its own separate pump, **AND**
 - **continue sodium chloride 0.9% solution as clinically indicated.**
 - **Patients must always continue their usual dose of subcutaneous long-acting basal insulin – prescribe on the Subcutaneous Insulin Order and Blood Glucose Monitoring Chart.**

9.0 Monitoring

- Ensure that clinical and biochemical parameters are improving.
- Check the following every hour for the first 6 hours, then at least 2 hourly if satisfactory response to interventions:
 - Repeat electrolytes (e.g. plasma sodium, potassium), urea and blood glucose testing.
 - Calculate serum osmolality = $[2(\text{sodium}) + \text{glucose} + \text{urea}]$ **and plot on the osmolality graph on MR 848** (see [Figure 1](#) for image).
 - Assess for complications of treatment e.g. fluid overload, cerebral oedema or central pontine myelinolysis (as indicated by a deteriorating conscious level).

Figure 1: Osmolality graph image



- If osmolality is increasing (or falling at a rate of $< 3\text{mOsmol/kg/hr}$) **and** sodium is increasing, check fluid balance:
 - If inadequate, increase the sodium chloride 0.9% infusion rate.
 - If adequate, consider changing to sodium chloride 0.45% infused at the same rate.
- If osmolality is falling at a rate $> 8\text{mOsmol/kg/hr}$, consider:
 - Reducing the rate of IV fluids.
 - Reducing the rate of insulin infusion (if commenced).

At all times, if the patient is not improving, senior clinician advice must be sought.

10.0 Other Considerations

- Identify and treat the underlying precipitant - careful examination and screening for underlying causes, and regular monitoring and re-evaluation is necessary.
- Consider underlying causes such as sepsis (including lower limb cellulitis, meningitis); silent ischaemia; and toxins (alcohol).
- If infection is present refer to the [Therapeutic Guidelines](#) or [Statewide Medicines Formulary](#).
- **Commence prophylactic anticoagulation as the patient will have a high risk of venous thromboembolism.**
- Assess foot risk score – assume high risk if the patient is obtunded or uncooperative:
 - Ensure heels are off-loaded.
 - Ensure daily foot checks.

11.0 Ongoing Management

- Convert patient to subcutaneous insulin when biochemically stable and the patient has resumed eating and drinking normally.
- Transfer from IV insulin to subcutaneous insulin using 75% of the previous day's (24 hours) intravenous requirements. Prescribe on the Subcutaneous Insulin Order and Blood Glucose Monitoring Chart.
- The starting insulin can be a pre-mix analogue (e.g. NovoMix 30®) or basal insulin analogues (e.g. Optisulin® or Toujeo®).
- Patients with pre-existing insulin-treated Diabetes Mellitus should be re-started on their usual insulin regimen when clinically appropriate.
- Contact the diabetes team for education and follow up at patients discharge.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGOPHCG Policy High Risk Medications](#)
- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- NSQHS Standards: 1, 4

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No. 51 Medication Management.](#)

- [SCGH Nursing Practice Guideline No. 4 Intravenous Therapy.](#)
- [SCGH Nursing Practice Guideline No. 2 Diabetes Patient Management.](#)
- [SCGH Guideline HRM025 Adult Diabetic Ketoacidosis \(DKA\)](#)
- [SCGOPHCG Guideline HRM003 Insulin: Adult Variable Rate Intravenous](#)
- Adult Hyperosmolar Hyperglycaemic State Guideline and Management Record (MR 848) SCGH
- Subcutaneous Insulin Order and Blood Glucose Monitoring Chart.
- [SCGOPHCG Guideline MMG037 SGLT2 inhibitors \(gliflozins\) and euglycaemic ketoacidosis Guideline SCGH.](#)
- [KEMH Obstetrics and gynaecology CPG Diabetes](#)
- [PCH Emergency Department Guidelines Diabetic Ketoacidosis](#)

References

1. The management of the hyperosmolar hyperglycaemic state (HHS) in adults with diabetes. London, Joint British Diabetes Societies Inpatient Care Group. 2012. Retrieved from: https://abcd.care/sites/abcd.care/files/resources/JBDS_IP_HHS_Adults.pdf

Authorisation & Version Control

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Applicable NSQHS Standards:	 Clinical Governance ☒	 Partnering with Consumers ☐	 Preventing and Controlling Healthcare Associated Infections ☐	 Medication Safety ☒	 Comprehensive Care ☐	 Communicating for Safety ☐	 Blood Management ☐	 Recognising and Responding to Acute Deterioration ☐
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Document Version Control								
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Appendix 1. Adult HHS Management Record

SIR CHARLES GAIRDNER HOSPITAL OSBORNE PARK HEALTH CARE GROUP ADULT HYPEROSMOLAR HYPERGLYCAEMIC STATE (HHS – FORMERLY KNOWN AS HONK) GUIDELINES AND MANAGEMENT RECORD		USE PATIENT LABEL WHEN AVAILABLE													
WARD: _____		Family name: _____	URN: _____												
CONSULTANT: _____		Given name(s): _____	DOB: _____												
NOTE: If Type 1 diabetes OR hyperglycaemic (blood glucose level >11mmol/L) with; Acidosis (venous bicarbonate <18mmol/L and/or pH < 7.35) AND blood (capillary) ketones (≥3 mmol/L) THEN FOLLOW DKA GUIDELINE MR836 & HRM025 These guidelines are for the treatment of adults with Hyperosmolar Hyperglycaemic State (HHS), previously known as hyperosmolar non-ketotic coma (HONK) and are designed to allow both prescriptions and relevant observations to be recorded together (refer to related guideline HRM026). The guideline should be adapted to the clinical situation where necessary. Seek advice from senior clinician/diabetes specialty units in cases of uncertainty. Continuous cardiac monitoring must be instituted when intravenous potassium chloride is administered at a rate of greater than 10mmol/hour. OPH patients diagnosed with DKA/HHS require early liaison with SCGH Endocrinology. Routine continuous cardiac monitoring for DKA / HHS patients is not available at OPH per the OPH High Risk Medicines Policy. Patients requiring intravenous potassium administered at a rate greater than 10mmol/hour must be transferred to SCGH. This guideline must not be used: • in children (<16years) even if they have HHS, seek specialist advice. OR • In the management of Diabetic Ketoacidosis (DKA) see MR849															
ENTRY CRITERIA: (Tick boxes if criteria present, if none re-consider diagnosis). Does patient have characteristic features of: • Hypovolaemia ☐ • Marked Hyperglycaemia (>30 mmol/L) ☐ • Increased serum/plasma osmolality (>320 mosmol/kg) [2(Sodium)+glucose+urea in mmol] ☐ If the patient does not meet any of the above criteria, re-consider diagnosis.															
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